
ElectViz Election Data Visualization for Media

Presented by
Vitta Hem Kumar



Introduction



The National Election Data Analysis Dashboard offers an insightful overview of India's electoral performance, highlighting how citizens vote and how parties perform across states. It visualizes key metrics such as total votes, vote shares, seat distribution, and party performance through interactive charts and maps. The dashboard enables analysis at both national and state levels, featuring major parties like BJP, Congress, AAP, and Telugu Desam Party. It also includes drill-down insights to the candidate level, showing winners, runners-up, and vote shares. Overall, it transforms complex election data into clear, meaningful insights that reflect India's democratic trends.



About Indian Election

What Is an Election?

- An election is a formal process through which people choose individuals to hold public office or make decisions on laws and policies. It's the cornerstone of democracy, giving citizens a voice in how they are governed.

Main Purposes of Elections

- Representation
- Accountability
- Legitimacy
- Participation

Types of Elections

- General Elections
- By-Elections
- Local Elections
- Referendums/Plebiscites

Important Terms

- Voter Turnout (%)
- Majority
- Margin of Victory
- Constituency
- Election Commission

Key Steps in an Election Process

- Announcement of Election Schedule
- Filing of Nominations by candidates or parties
- Scrutiny and Withdrawal of nominations
- Campaigning Period for political parties and candidates
- Polling Day – Voters cast ballots (paper or electronic)
- Counting of Votes
- Declaration of Results and Formation of Government

Dataset Overview

candidates_with_phase

- Contains personal and nomination details of each candidate.
- Includes fields like Candidate Name, Age, Gender, Address, Constituency, Application Date, and Application Status.

results_2024_winners

- Contains final election outcomes for each constituency.
- Fields include Winning Candidate, Winning Party, Runner-up Candidate, Runner-up Party, State, PC No, Margin Votes, and Result Status.

results_2024

- Stores detailed election data for all candidates and constituencies.
- Includes Candidate Name, Party, State, PC Name, PC No, EVM Votes, Postal Votes, and Total Votes.

indian-state-level-election

- The dataset contains election data from all Indian states for the year 1985.
- It includes details like state name, candidate info, party name, votes polled, and total electors.



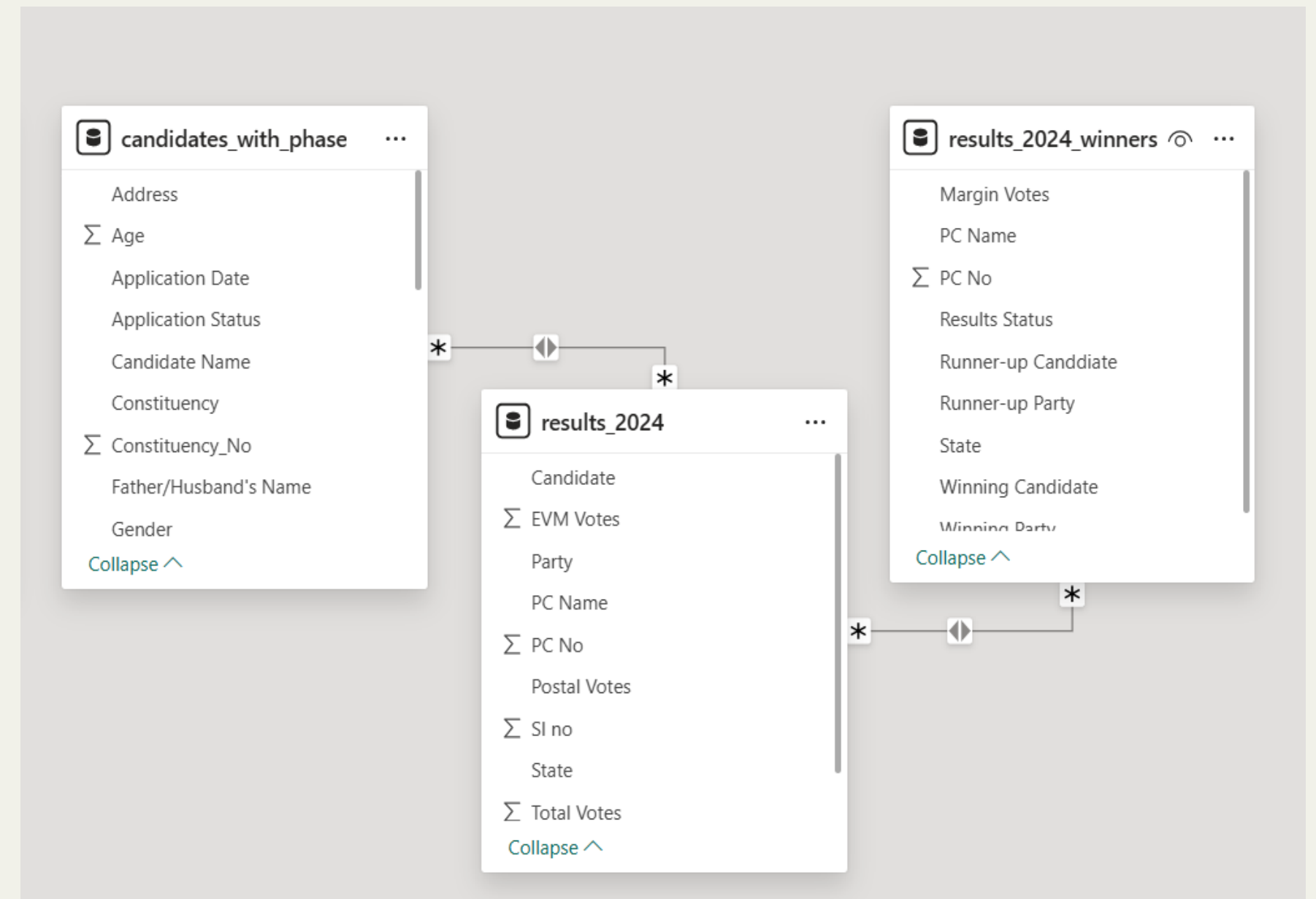
st_name ▾	year ▾	ac_no ▾	ac_name ▾	ac_type ▾	cand_name ▾	cand_sex ▾	partyname ▾	partyabbre ▾	totvotpoll ▾	electors ▾
Maharashtra	1985	1		GEN	Balkrishna Raghoba Sawant	M	Independent	IND	1534	111887
Maharashtra	1985	1		GEN	Vijay Kerkar	M	Independent	IND	1776	111887
Maharashtra	1985	1		GEN	Vasant Kesarkar	M	Independent	IND	6276	111887
Maharashtra	1985	2		GEN	Vilas Shantaram Tari	M	Independent	IND	246	104559
Maharashtra	1985	2		GEN	Dattatray Narayan Kamat	M	Independent	IND	307	104559
Maharashtra	1985	2		GEN	Narayan Vasudeo Pitre	M	Independent	IND	670	104559
Maharashtra	1985	2		GEN	C. S. Gaonkar	M	Independent	IND	858	104559
Maharashtra	1985	2		GEN	Vasantrao Khot	M	Independent	IND	1786	104559
Maharashtra	1985	2		GEN	Jagannath Ramachandra Ghatwal	M	Independent	IND	2682	104559
Maharashtra	1985	2		GEN	S. N. Desai	M	Independent	IND	8736	104559
Maharashtra	1985	3		GEN	Amrut Raghunath Parab	M	Independent	IND	175	100743
Maharashtra	1985	3		GEN	Vasantrao Dhanaji Moperkar	M	Independent	IND	189	100743
Maharashtra	1985	3		GEN	Bhai Chindarkar	M	Independent	IND	791	100743
Maharashtra	1985	3		GEN	Appasaheb Chavan	M	Independent	IND	2331	100743
Maharashtra	1985	3		GEN	Kalabhai Vithoba Asant	M	Independent	IND	3171	100743

PC Name ▾	Winning Candidate ▾	Winning Party ▾	Runner-up Canddiate ▾	Runner-up Party ▾	Margin Vote:
Arunachal East	TAPIR GAO	Bharatiya Janata Party	BOSIRAM SIRAM	Indian National Congress	30421
Arunachal West	KIREN RIJUJU	Bharatiya Janata Party	NABAM TUKI	Indian National Congress	100738
Darrang-Udalguri	DILIP SAIKIA	Bharatiya Janata Party	MADHAB RAJBANGSHI	Indian National Congress	329012
Guwahati	BIJULI KALITA MEDHI	Bharatiya Janata Party	MIRA BORTHAKUR GOSWAMI	Indian National Congress	251090
Karimganj	KRIPANATH MALLAH	Bharatiya Janata Party	HAFIZ RASHID AHMED CHOUDHURY	Indian National Congress	18360
Kaziranga	KAMAKHYA PRASAD TASA	Bharatiya Janata Party	ROSELINA TIRKEY	Indian National Congress	248947
Lakhimpur	PRADAN BARUAH	Bharatiya Janata Party	UDAY SHANKAR HAZARIKA	Indian National Congress	201257
Silchar	PARIMAL SUKLABAIDYA	Bharatiya Janata Party	SURYA KANTA SARKAR	Indian National Congress	264311
Sonitpur	RANJIT DUTTA	Bharatiya Janata Party	PREMLAL GANJU	Indian National Congress	361408
Maharajganj	JANARDAN SINGH "SIGRIWAL"	Bharatiya Janata Party	AAKASH KUMAR SINGH	Indian National Congress	102651
Muzaffarpur	RAJ BHUSHAN CHOUDHARY	Bharatiya Janata Party	AJAY NISHAD	Indian National Congress	234927
Paschim Champaran	DR.SANJAY JAISWAL	Bharatiya Janata Party	MADAN MOHAN TIWARI	Indian National Congress	136568
Patna Sahib	RAVI SHANKAR PRASAD	Bharatiya Janata Party	ANSHUL AVIJIT	Indian National Congress	153846

Dataset Schema

The data schema defines the structured relationship between the datasets used in the Election Analysis Dashboard. It integrates three key tables — candidates_with_phase, results_2024, and results_2024_winners — ensuring accurate and seamless data flow for analysis.

Relationships are built through the Constituency Number (PC No) and Candidate Name fields, enabling cross-table analysis. This schema design allows efficient data linking for visualizing candidate-level, party-level, and state-level insights in the dashboard.



Data Preparation

Data preparation involved collecting, cleaning, and organizing election datasets from multiple sources to ensure accuracy and consistency. The processed data was then structured for visualization, enabling efficient analysis of votes, seats, and party performance.

Steps

Data Cleaning

- Removed duplicates,
- Handled missing values,
- Handling datatypes

Data Transformation

- Added custom columns turnout rate, Winning party, Swingflag columns

Custom Queries

- Age Group
- Region Type (Urban or Rural)
- Candidate Vote Share (Vote Share)

DAX in Power BI

- DAX (Data Analysis Expressions) is a formula language used in Power BI.
- To perform calculations, create measures, and build calculated columns.

How DAX Helps in Power BI :-

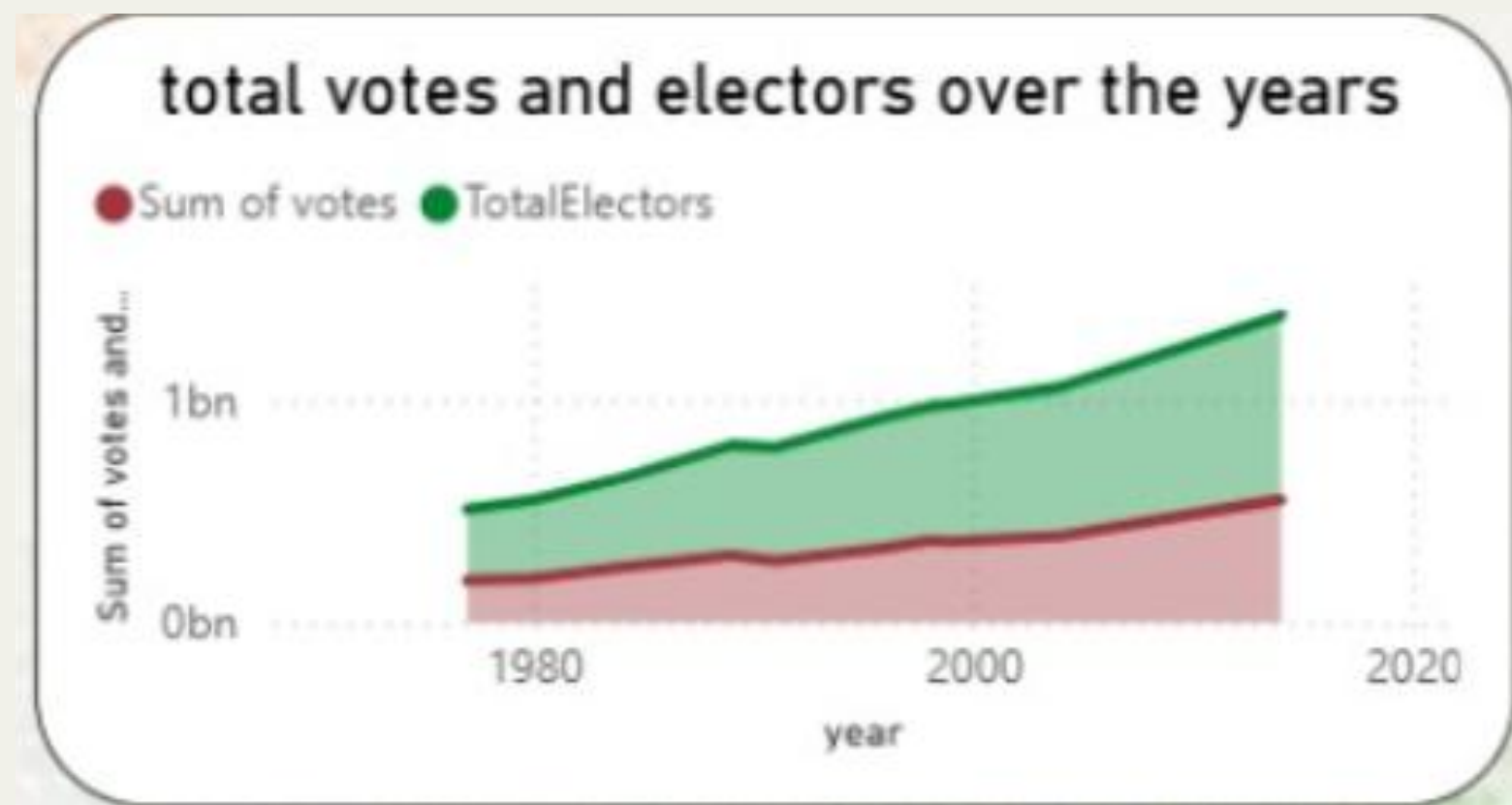
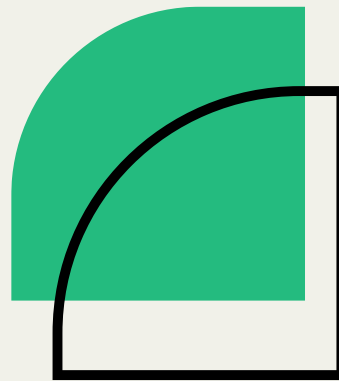
Advanced Analysis, Dynamic Reporting, Custom KPIs

Measure in Power BI

- A Measure in Power BI is a calculation used to analyze data dynamically.
- It's write using DAX (Data Analysis Expressions) and performs computations like totals, averages, percentages, ratios, or KPIs.

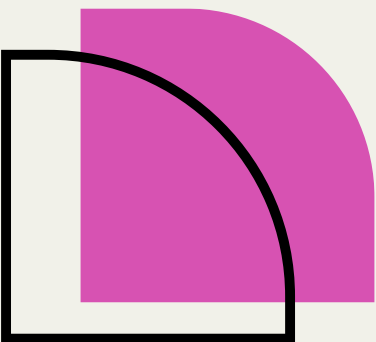
Example :-

1) **APP % of Seats** = `DIVIDE([AAP_seats], [total_seats], 0)`
2) **total_seats** = `CALCULATE(COUNT(results_2024_winners[Winning Party]))`



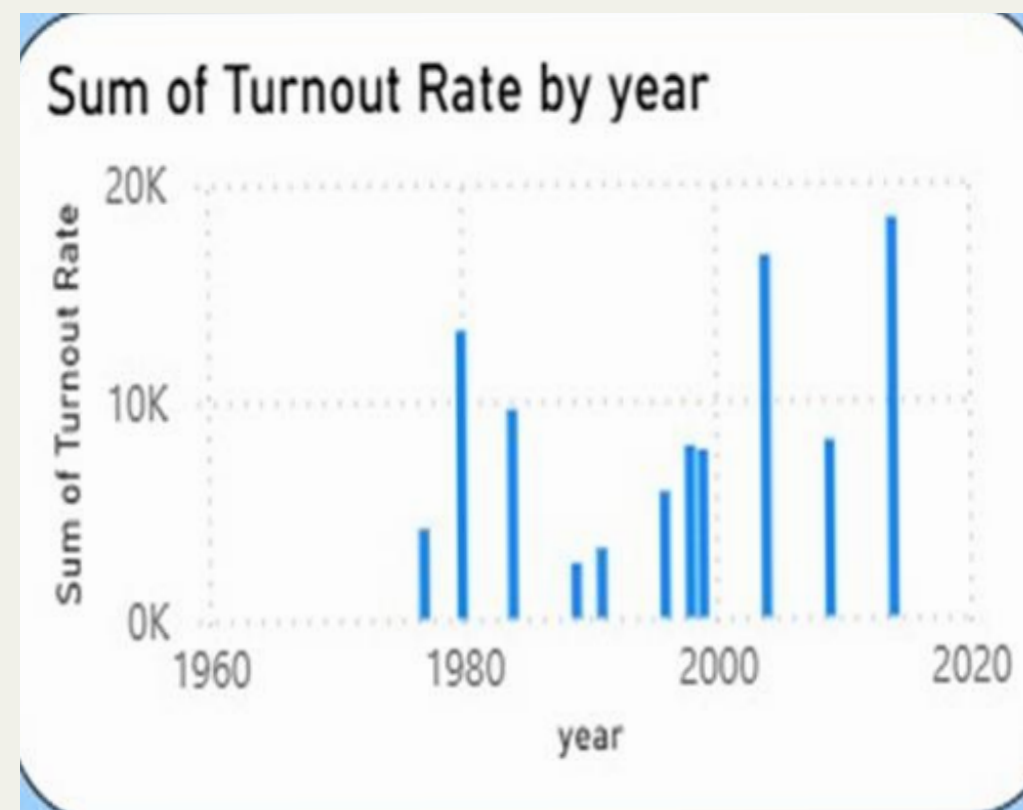
Historical Trends

- Show patterns and changes in election outcomes over different years.
- Help identify shifts in party dominance and voter preferences across regions.
- Allow comparison of vote shares, seat counts, and turnout rates over time.
- Reveal long-term political movements and the rise or decline of major parties.
- Support prediction and analysis of future electoral behavior based on past data.



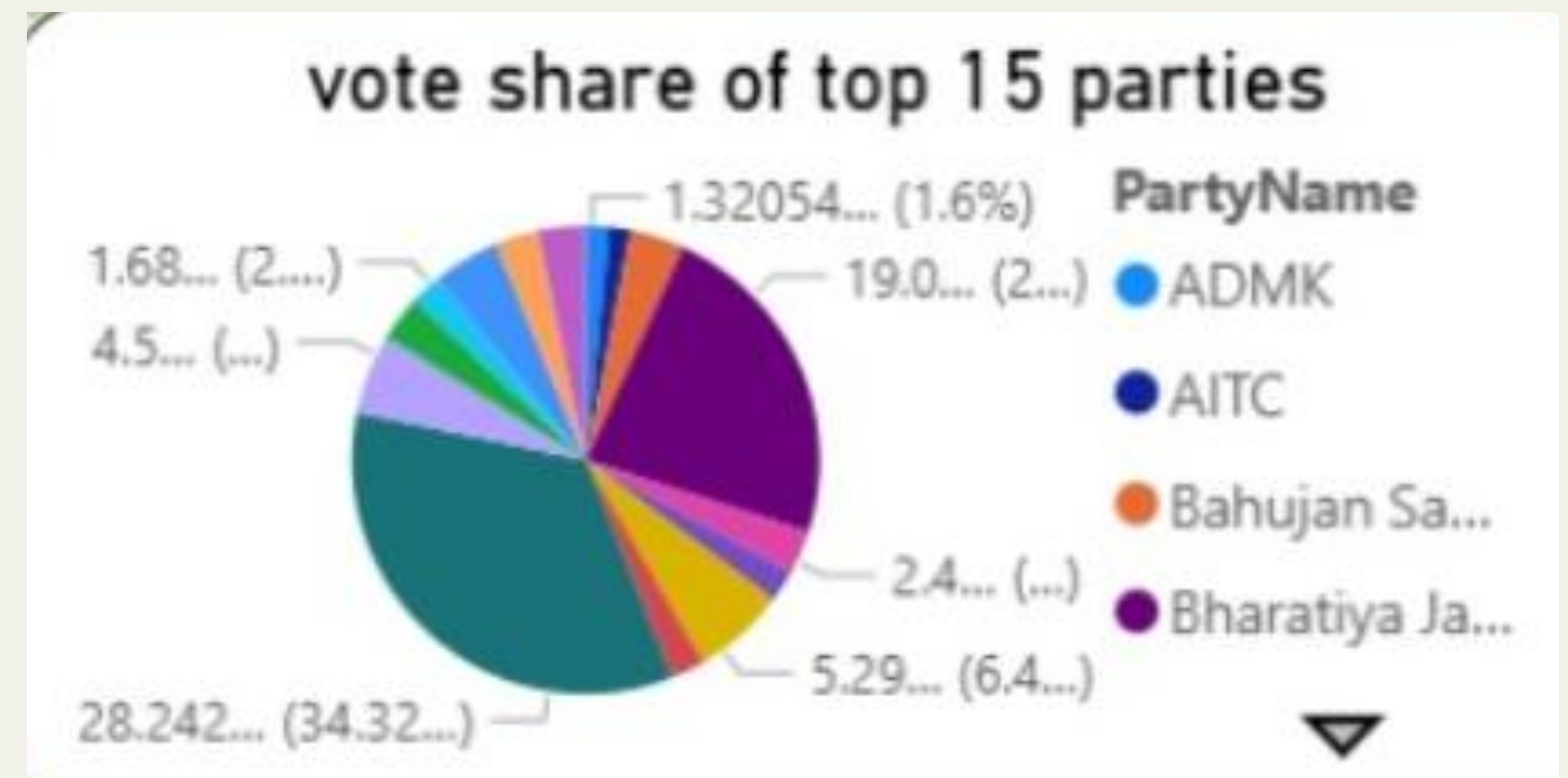
Voter Turnout

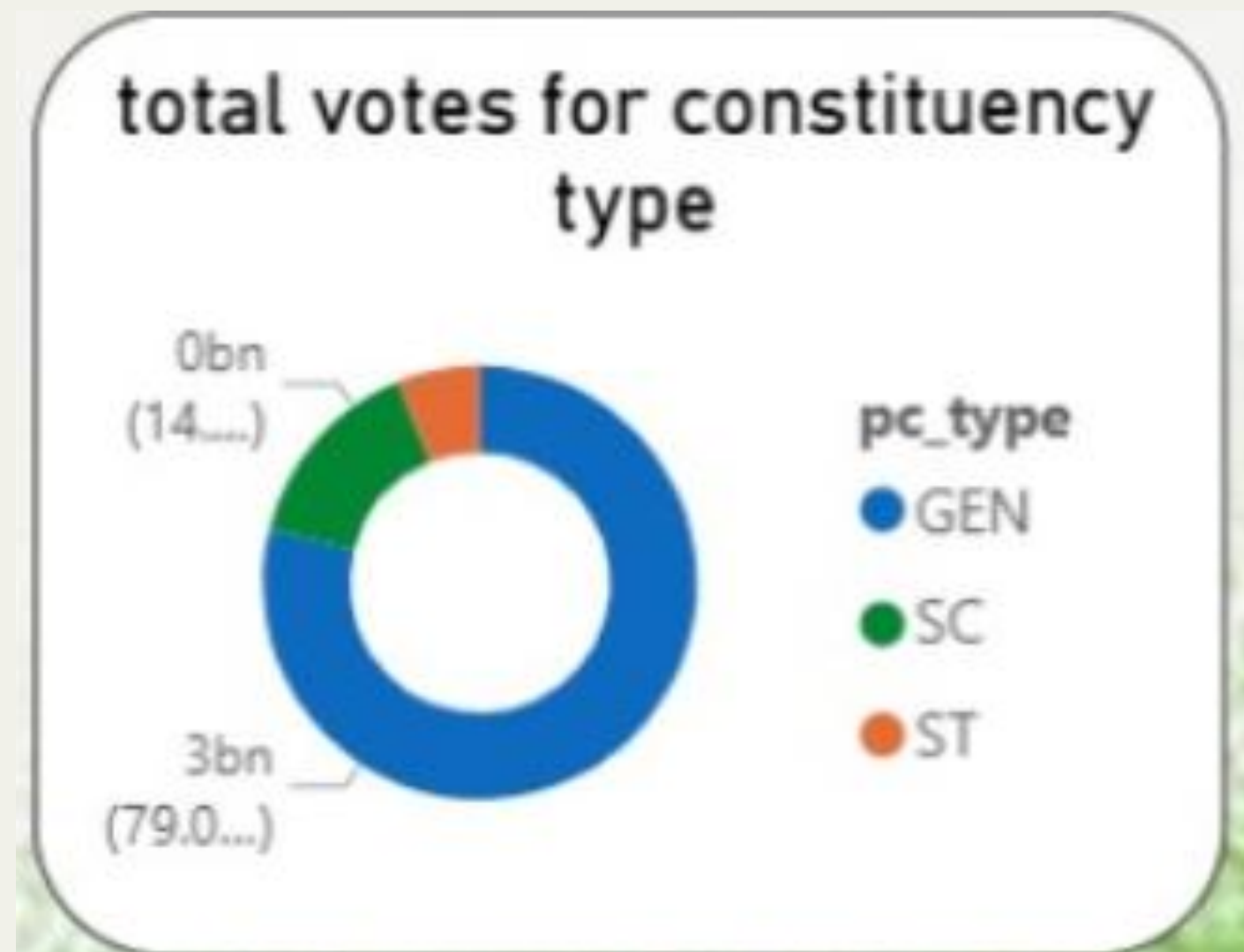
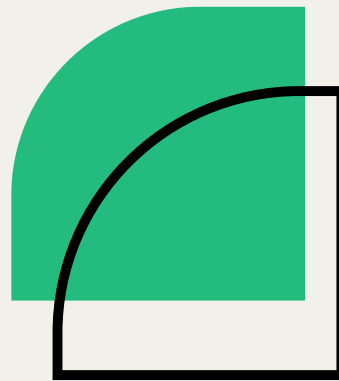
- Represents the percentage of eligible citizens who cast their vote, indicating the level of public participation in elections.
- Indicates the level of public participation and political engagement in the democratic process.



Vote Share

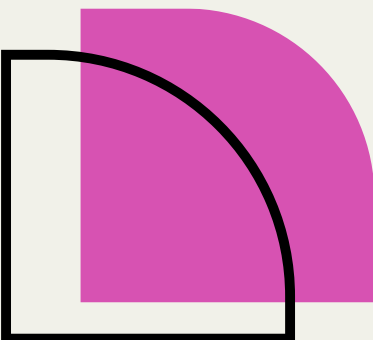
- Shows the percentage of total votes secured by a party or candidate, highlighting their electoral strength.
- Together, turnout and vote share reveal both citizen engagement and political performance across elections.





Swing Constituencies and Long-Term Strongholds

- Swing Constituencies are regions where election outcomes frequently change between parties, reflecting dynamic voter behavior.
- They play a crucial role in determining overall results due to their unpredictable nature.
- Long-Term Strongholds are areas consistently dominated by a single party across multiple elections.
- Such regions indicate strong party loyalty, stable voter support, and effective local leadership.
- Analyzing both helps understand political stability and shifting voter trends across regions.



Party Trends

```
graph TD; A[Party Trends] --> B[Show how major political parties perform across states and over different election years]; A --> C[Highlight changes in vote share, seat distribution, and regional influence]; A --> D[Reveal emerging parties, shifting voter preferences, and evolving political dynamics];
```

Show how major political parties perform across states and over different election years

Highlight changes in vote share, seat distribution, and regional influence

Reveal emerging parties, shifting voter preferences, and evolving political dynamics

What is API ?

Definition

- An API (Application Programming Interface) enables communication between different software systems.
- It defines how data requests and responses are structured and exchanged.
- In this project, it acts as a medium to connect the election data source with the visualization dashboard.

Role

- Fetches real-time election data such as vote counts, party standings, and constituency updates.
- Ensures smooth data flow between backend systems and the dashboard.
- Allows automatic data refresh without manual input.
- Enhances data accuracy and user engagement through live updates.

Type

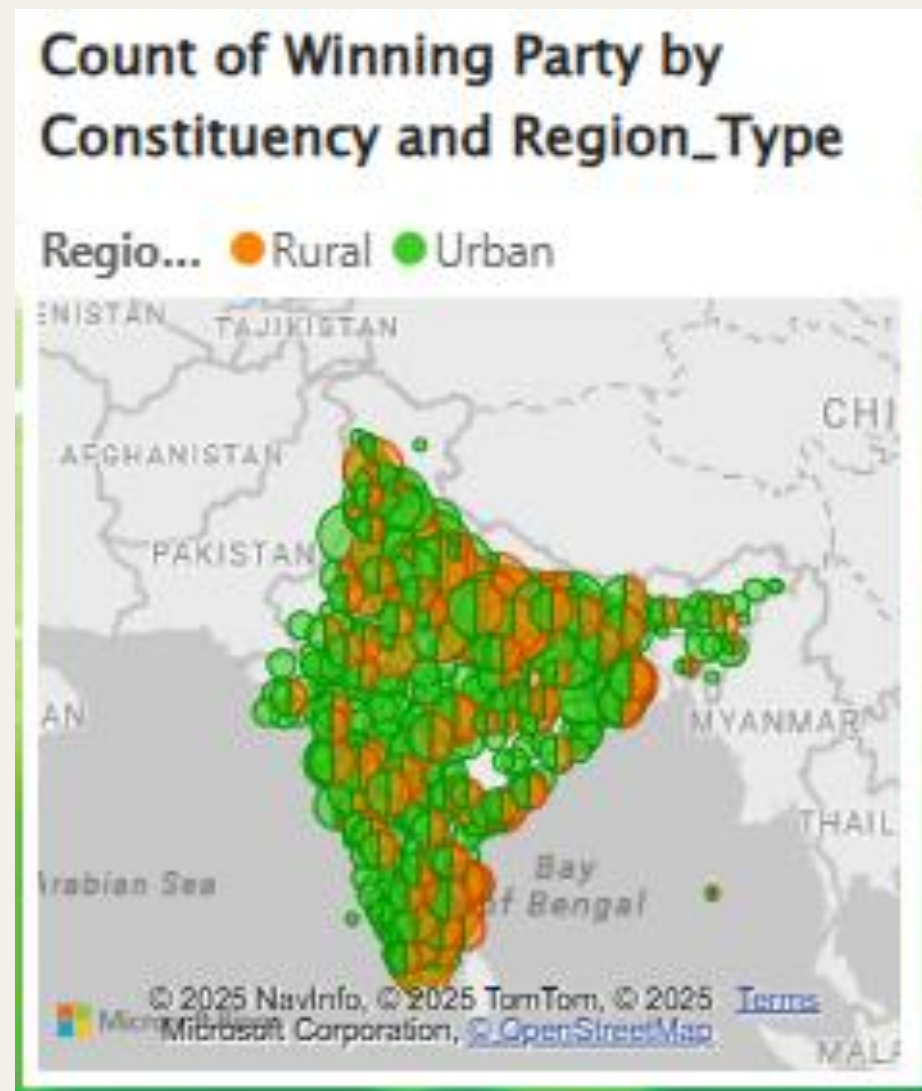
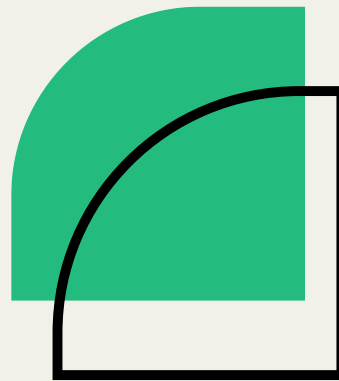
- REST API is used for efficient communication using HTTP methods (GET, POST, PUT, DELETE).
- Provides structured JSON data compatible with visualization tools.
- Mock API is used for simulation during testing or when real-time data isn't available.
- Enables developers to test and visualize results before actual data integration

Demographic Analysis



How to assess your leadership style

- ✓ Identifies voting trends among specific groups such as youth, women, and first-time voters.
- ✓ Highlights regional diversity and its impact on political outcomes.
- ✓ Helps parties tailor campaigns and policies to target key voter segments.
- ✓ Shows the relationship between literacy, income levels, and voter participation.
- ✓ Aids in understanding socio-economic and cultural influences on voting behavior.

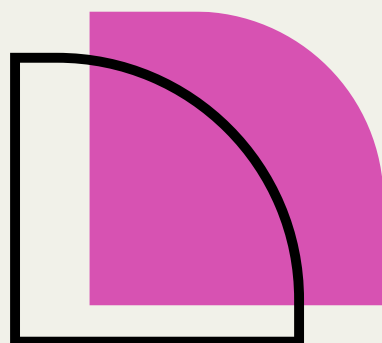


Urban vs Rural

- Compare candidate vote share across Urban and Rural regions
- Identify which regions favor specific candidates or demographics
- Helps understand regional voting patterns and trends

We created a custom column named Region Type that classifies candidates into Urban or Rural based on their constituency address. This enables its use in slicers, filters, and visuals for dynamic analysis.

Region Type =
IF(
SEARCH("Village", 'candidates_with_phase'[Address], 1, 0) > 0
|| SEARCH("Taluk", 'candidates_with_phase'[Address], 1, 0) > 0,
"Rural",
"Urban"
)





Drill-down Analysis



Map

- Displays region-wise winning parties and vote distributions through interactive visuals.
- Helps identify political strongholds, swing regions, and regional voting patterns.
- Enhances understanding of spatial trends in voter behavior and party dominance.

Matrix

- Provides detailed numerical insights such as total votes, seat count, and vote share percentages.
- Enables comparison of party performance across different years and constituencies.
- Includes candidate-level data like winners, runners-up, and margin of victory.



INDIAN STATE ELECTION ANALYSIS



Election Year

All

Political Party

All

Select State

All

Constituency Type

All

3044M

Total Votes

1637

Total Parties

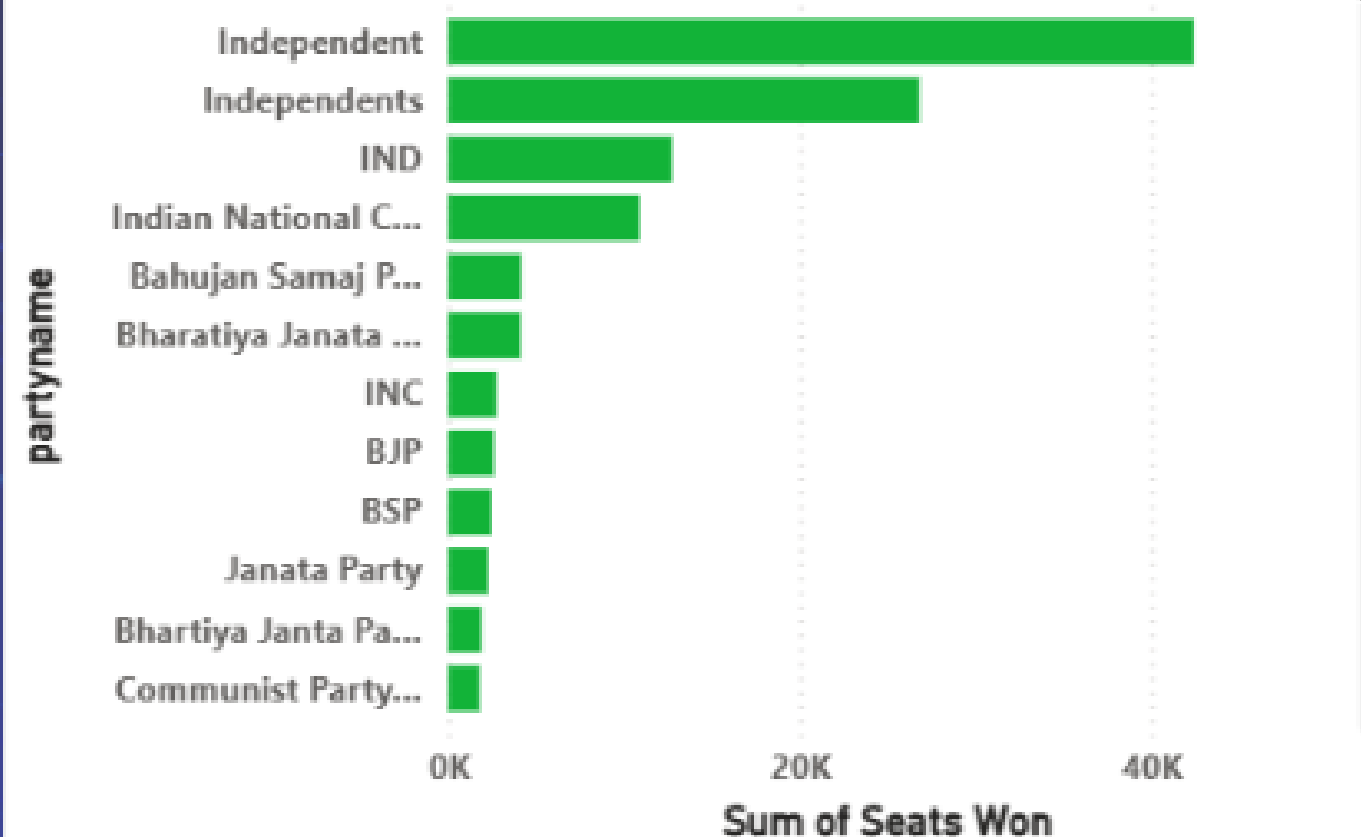
5670

Total Seats

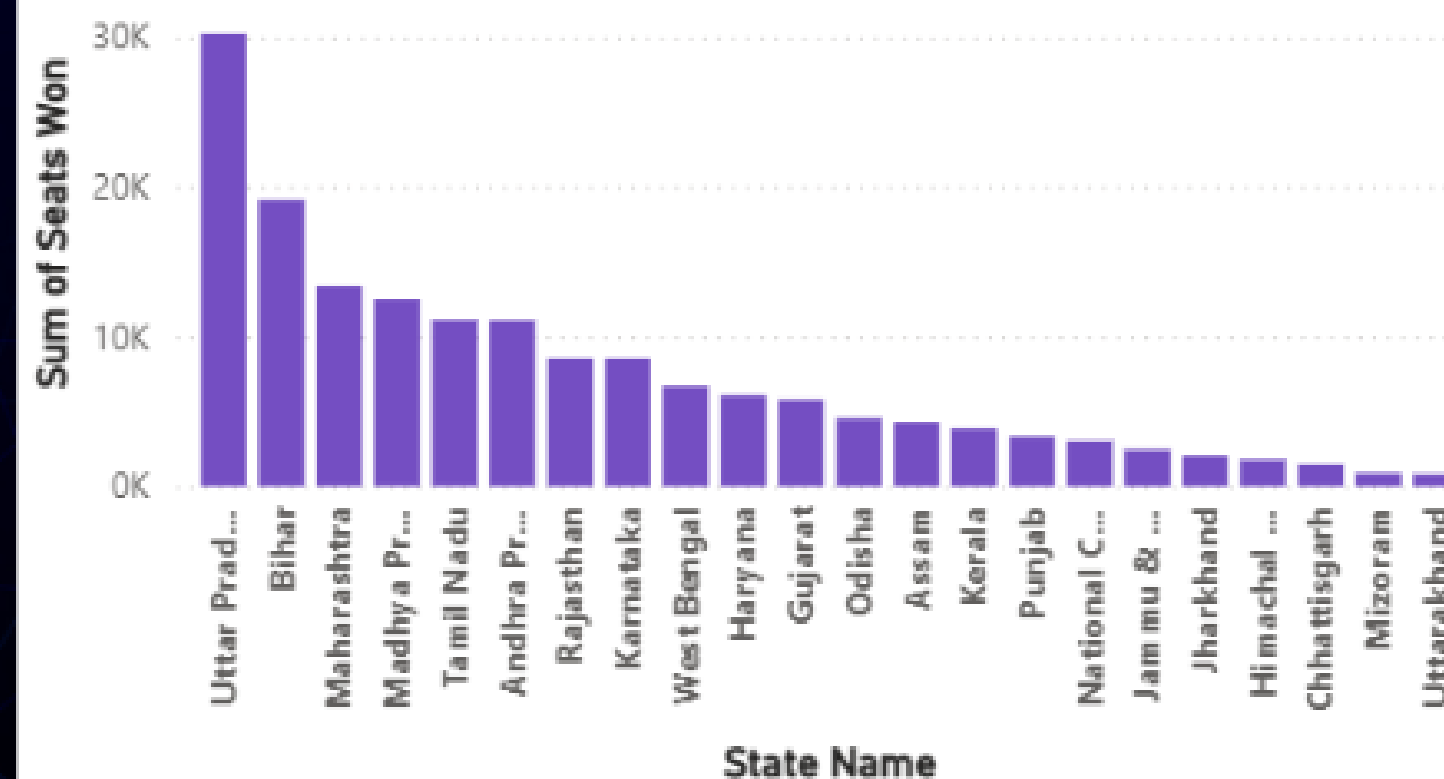
5.52

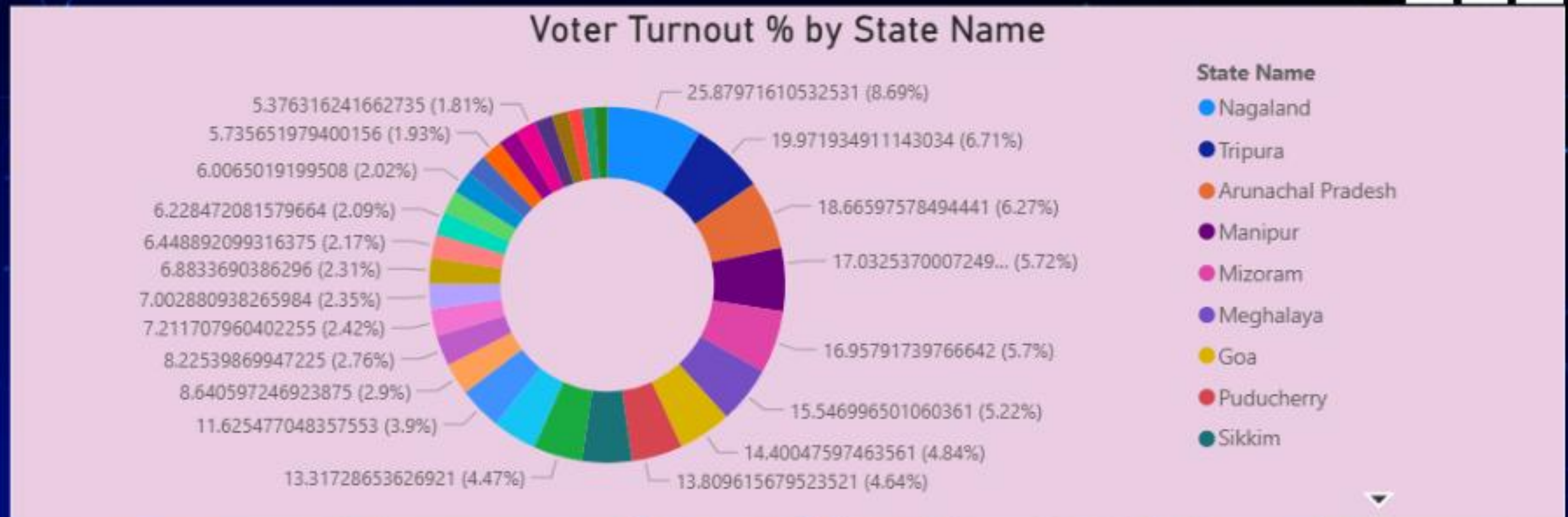
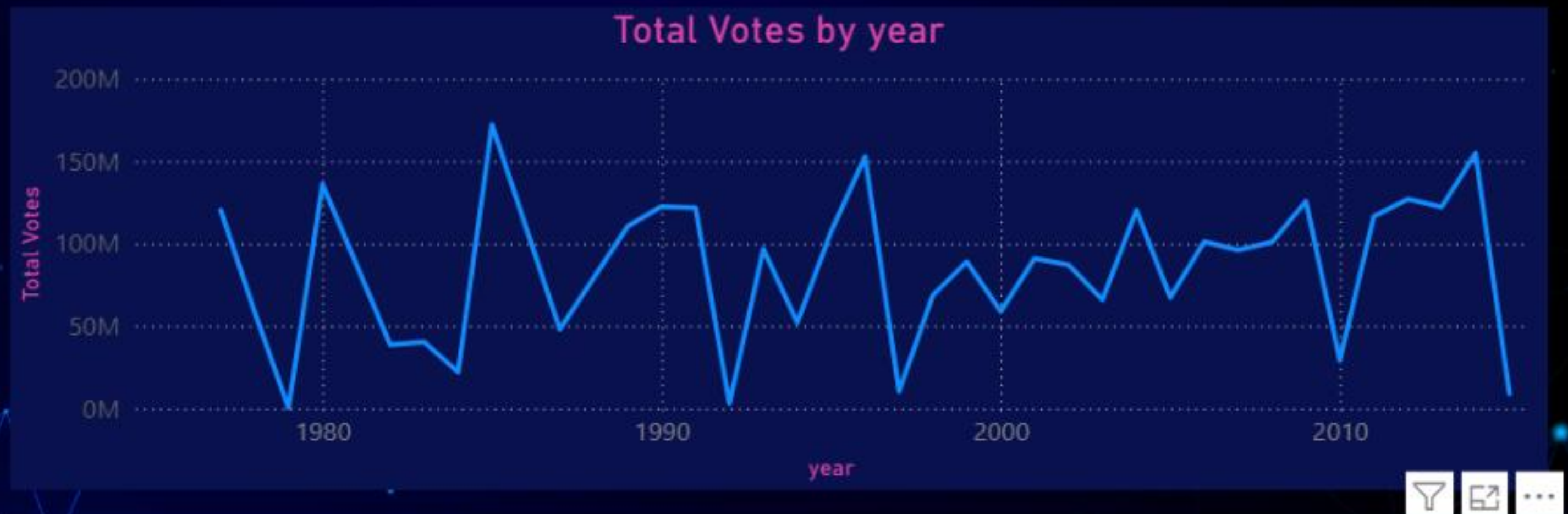
Voter Turnout %

Seats Won by party Name



Sum of Seats Won by State





Top Party

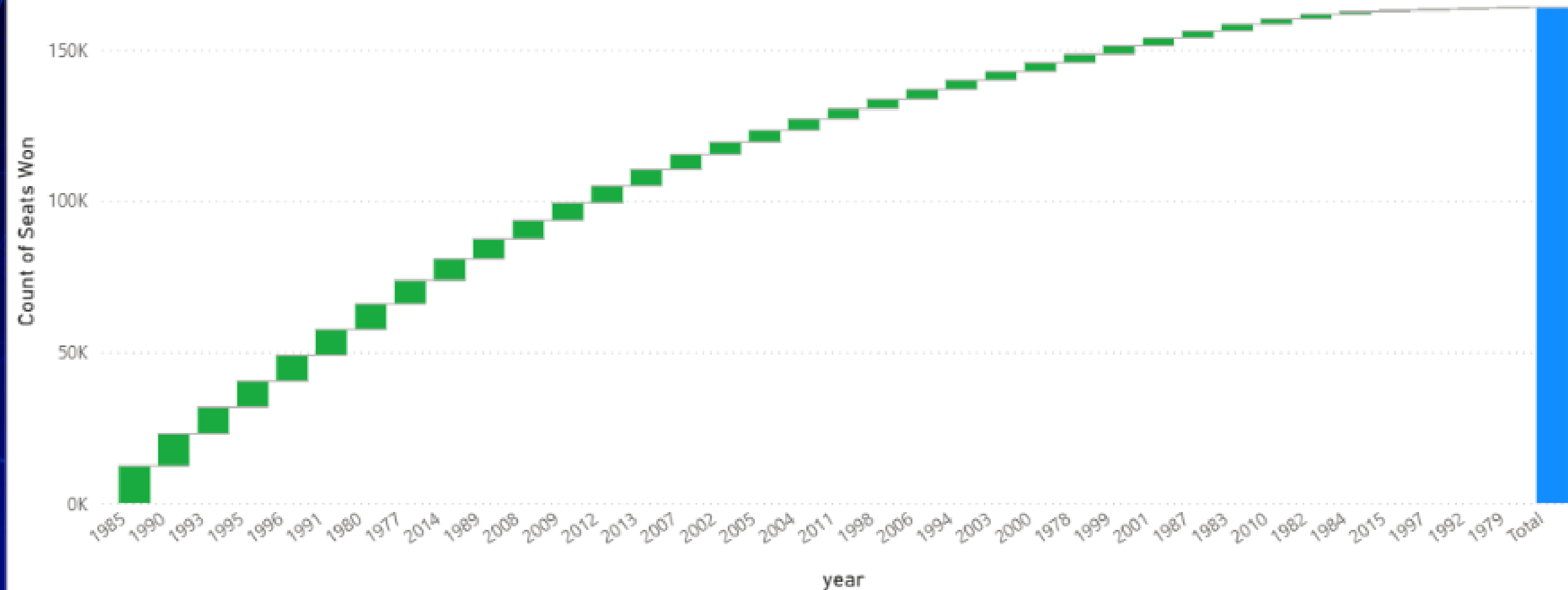
Independent

Top State

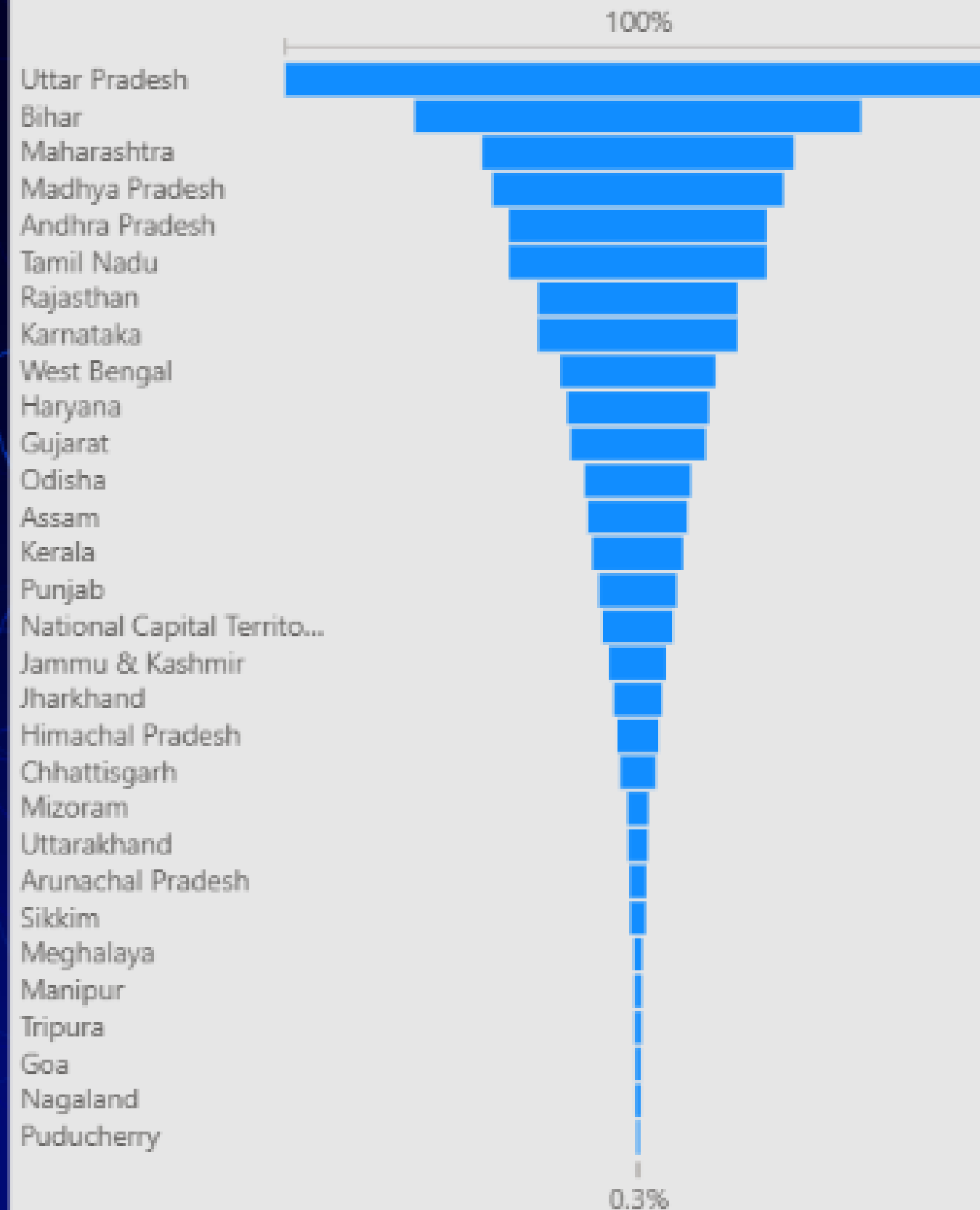
Uttar Pradesh

Count of Seats Won by year

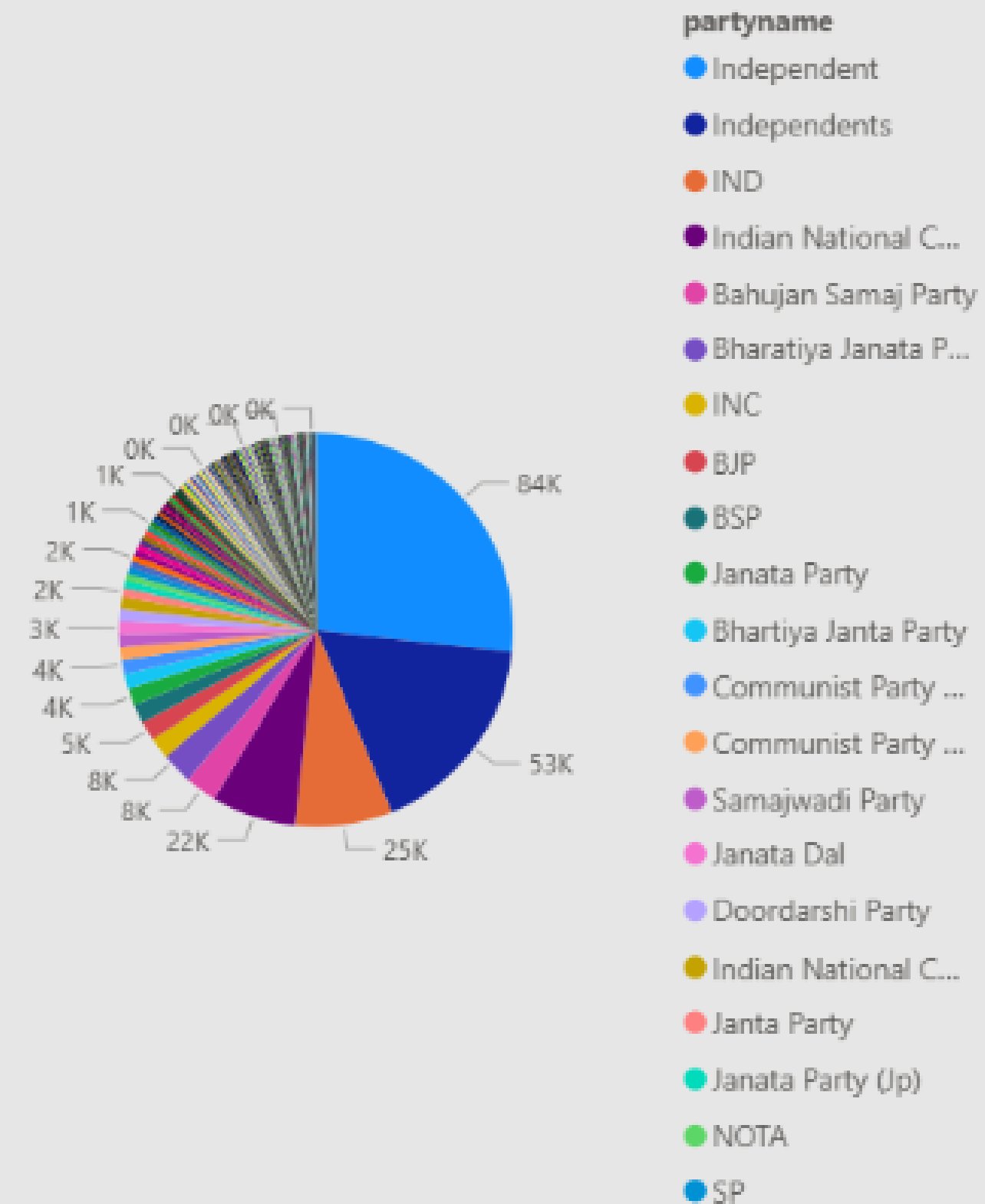
● Increase ● Decrease ● Total



Count of Seats Won by State name



Vote share by party



State-wise Party Seat Distribution

State Name	A S P	Aa S P	AAAP	AACP	Aadivasi Sena Party	AAMP	AAP	AASAP	AaSP	ABAS	ABCD(A)	ABDBM	ABGL	ABGP	ABHM	AB
Andhra Pradesh			79													
Arunachal Pradesh																
Assam																
Bihar																
Chhattisgarh		1												2	1	
Goa																
Gujarat					2											
Haryana																
Himachal Pradesh																
Jammu & Kashmir																
Jharkhand						1									2	
Karnataka															1	
Kerala																
Madhya Pradesh									8					16	3	
Maharashtra															11	
Manipur																
Meghalaya																
Mizoram																
Nagaland																
National Capital Territory Of Delhi		2	59	4				1							5	
Odisha			57													
Puducherry																
Punjab																
Rajasthan							2				4				11	
Sikkim																
Tamil Nadu															6	
Tripura																
Uttar Pradesh	4	1								1		1			6	
Total	4	4	195	4	2	1	2	1	8	1	4	1	2	18	47	



Thank You